

Practice Quiz 03C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 03

Nested quantifiers and negating quantified statements.

1. Consider

$$(i) \forall n \in \mathbf{N} \exists m \in \mathbf{N} (m = n + 1) \quad (ii) \exists m \in \mathbf{N} \forall n \in \mathbf{N} (m = n + 1).$$

(a) State each in plain English and say whether it is true or false.

Solution Statement (i) says that every natural number has a natural-number successor. It is true: take $m = n + 1$. Statement (ii) says that one fixed natural number is the successor of every natural number. It is false. ■

(b) Negate (ii), with no negation symbol to the left of a quantifier.

Solution Its negation is

$$\forall m \in \mathbf{N} \exists n \in \mathbf{N} (m \neq n + 1).$$

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